

Deep Learning Project 3 plan Image generation

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1 Introduction

Generative deep learning models have recently achieved remarkable success in image synthesis tasks. Architectures such as Generative Adversarial Networks (GANs) [3], Variational Autoencoders (VAEs) [8], and their modern extensions enable the generation of realistic high-dimensional images by learning the underlying probability distribution of training data.

The objective of this project is to investigate deep generative models for unconditional image synthesis using a dataset of cat images [2]. We will conduct a comparative study of two fundamentally different generative paradigms: adversarial training and latent-variable autoencoding. In particular, the project will compare a Wasserstein GAN with Gradient Penalty (WGAN-GP) [4] and a Vector Quantized Variational Autoencoder (VQ-VAE) [10].

Our analysis will focus on generation quality, training stability, latent space structure, and computational efficiency under limited hardware resources. Special attention will be devoted to mode collapse phenomena, latent interpolation behavior, and the ability of the learned latent representations to encode semantically meaningful visual attributes.

2 Dataset

The experiments will mainly be conducted on the Cat Dataset. The dataset contains thousands of images of domestic cats collected under varying lighting conditions, poses, backgrounds, and image resolutions.

The original dataset contains images with inconsistent resolutions and aspect ratios, which requires a unified preprocessing pipeline before training generative models.

The preprocessing procedure will include:

- **Resolution Standardization:** All images will be resized to a fixed resolution of 64×64 pixels to reduce computational requirements and ensure architectural compatibility.
- **Aspect Ratio Preservation:** Images will first be center-cropped to a square aspect ratio before resizing, preventing geometric distortion of cat facial features and body structure.
- **Data Augmentation:** Random horizontal flips will be applied during training to improve generalization and increase effective dataset diversity without additional storage costs.
- **Normalization:** Pixel intensities will be scaled either to $[-1, 1]$ for adversarial training stability or to $[0, 1]$ for autoencoder-based frameworks.

Additionally, an exploratory extension of the project will utilize the Dogs vs. Cats [6] dataset to investigate whether unconditional generative models can naturally separate mixed semantic distributions.

3 Compared Network Architectures

In this study, three generative architectures of varying complexity and modeling paradigms will be evaluated:

- **Wasserstein GAN with Gradient Penalty (WGAN-GP)** [1, 4]: An adversarial generative model designed to improve training stability and reduce mode collapse through Wasserstein distance optimization and gradient penalty regularization.
- **Vector Quantized Variational Autoencoder (VQ-VAE)** [10]: A latent-variable generative model that utilizes discrete latent representations through vector quantization to generate sharper and more detailed images than standard VAEs.
- **StyleGAN2-ADA** [7]: A high-capacity GAN architecture with adaptive discriminator augmentation designed for stable training on limited datasets, used as a fine-tuned benchmark model.

For the WGAN-GP and VQ-VAE architectures, we prioritize training from scratch to facilitate an empirical analysis of architectural and optimization-related hyperparameters. Given our computational constraints, this will likely involve implementing reduced-scale variants of the models with lower image resolutions and simplified network depths.

The StyleGAN2-ADA architecture will primarily serve as a fine-tuned reference model and performance benchmark.

4 Experiments

4.1 Hyperparameter Optimization and Training Stability

The goal of this stage is to identify stable and well-performing configurations for each model before conducting any comparative evaluation.

WGAN-GP. We will vary two key optimization hyperparameters:

- **Gradient penalty coefficient** $\lambda \in \{5, 10, 15\}$: controls the strength of the Lipschitz constraint imposed on the critic.
- **Critic-to-generator update ratio** $n_{\text{critic}} \in \{2, 3, 4\}$: the number of critic gradient steps performed per generator step. A higher ratio promotes a more accurate Wasserstein estimate but increases training time.

VQ-VAE. We will vary two hyperparameters governing the discrete bottleneck:

- **Commitment loss coefficient** $\beta \in \{0.1, 0.5, 1.0\}$: balances the encoder commitment to the codebook against reconstruction fidelity.
- **Codebook size** $K \in \{256, 512, 1024\}$ (with fixed embedding dimension $D = 64$): determines the expressiveness of the discrete latent space. We will track codebook utilization (the fraction of active codes) to diagnose codebook collapse.

The best configuration for each model, will be carried forward to subsequent experiments.

4.2 Quantitative and Qualitative Comparative Evaluation

The second stage focuses on comparing the performance of the three generative paradigms.

- **Quantitative Evaluation:** The Fréchet Inception Distance (FID) [5] will be computed on a validation subset. Lower FID values indicate higher similarity between generated and real image distributions.
- **Qualitative Evaluation:** Generated images will be visually inspected with emphasis on:
 - Texture sharpness and fur detail,
 - Anatomical correctness of facial structures,
 - Presence of visual artifacts or repeated patterns.

This comparison will help determine the trade-offs between image realism, diversity, and training stability.

4.3 Linear Interpolation Experiment

This experiment investigates the geometric structure and smoothness of the learned latent space representations.

Two distinct latent vectors, z_A and z_B , generating visually different but high-quality cat images, will be selected. Linear interpolation will then be performed according to:

$$z_t = (1 - t)z_A + t \cdot z_B.$$

This process will generate eight intermediate latent representations, producing a total of ten images including the original endpoints.

The generated interpolation trajectory will be analyzed to determine whether the model learns a smooth semantic manifold or exhibits abrupt transitions and unstable latent regions.

4.4 Multi-Class Learning (Cats vs. Dogs)

This experiment investigates how unconditional generative models behave when trained on a mixed semantic distribution.

Setup. The best-performing architecture from Section 4.2 will be retrained on the combined Cats vs. Dogs dataset [6], applying the same preprocessing pipeline.

Research questions. We are particularly interested in which of the following behaviors the model exhibits:

- **Distribution separation:** the model implicitly partitions the latent space into cat-like and dog-like regions, producing recognizable samples from both classes.
- **Mode collapse to a dominant class:** the model ignores one class and generates samples exclusively from the other.
- **Semantic hybridization:** the model blends features of both classes, producing anatomically inconsistent “cat-dog” chimeras.

4.5 (Additional Experiment): Latent Vector Arithmetic

Inspired by semantic directions discovered in word embedding spaces [9] and the disentangled style spaces of StyleGAN, in this experiment we will test whether our unconditionally trained models encode semantically interpretable directions in their continuous latent spaces.

Methodology. The procedure requires no additional training and operates entirely on the frozen model weights:

1. Select a binary visual attribute X (e.g., “dark fur” or “open mouth”).
2. Manually identify 15 generated images that clearly display attribute X and 15 that do not, retrieving their corresponding latent vectors $\{\mathbf{z}_i^+\}$ and $\{\mathbf{z}_i^-\}$.
3. Compute the mean attribute direction:

$$\mathbf{d}_X = \frac{1}{15} \sum_{i=1}^{15} \mathbf{z}_i^+ - \frac{1}{15} \sum_{i=1}^{15} \mathbf{z}_i^-. \quad (1)$$

4. Apply the direction to a neutral image’s latent vector $\mathbf{z}_0$ by computing $\mathbf{z}_0 + \alpha \mathbf{d}_X$ for $\alpha \in \{-1, 0, 1\}$ and decoding the result.

Significance. A successful manipulation – in which the target attribute is added or removed while other visual properties remain unchanged – would demonstrate that the model has learned a disentangled, linearly manipulable latent representation of semantic features, despite having received no explicit supervision about those attributes.

5 Result Evaluation

The results of all models will be evaluated using two complementary approaches. The primary quantitative metric will be the **Fréchet Inception Distance (FID)**, computed between a large sample of generated images and the real validation set. FID captures both the fidelity and diversity of generated outputs, with lower values indicating higher generation quality. For the VQ-VAE model, **codebook utilization** will additionally be tracked to diagnose codebook collapse throughout training.

Quantitative scores will be supplemented by qualitative visual inspection of randomly sampled image grids. Generated outputs will be assessed for texture sharpness, anatomical plausibility of facial structures, diversity of poses and backgrounds, and the presence of visible artifacts.

6 Conclusion

This project will compare two fundamentally different generative paradigms – **WGAN-GP** and **VQ-VAE** – for unconditional image synthesis on a dataset of cat images. Additionally, **StyleGAN2-ADA** will be fine-tuned on the same data to serve as a high-capacity benchmark, allowing us to quantify the performance gap between models trained from scratch and a state-of-the-art pretrained architecture. The main goal is to analyse how architectural choices, optimization hyperparameters, and latent space structure affect generation quality and training stability.

The study will also investigate the geometric properties of the learned latent spaces through linear interpolation, and explore whether unconditionally trained models can naturally separate mixed semantic distributions in the Cats vs. Dogs setting. The additional latent vector arithmetic experiment will probe the degree of disentanglement achieved without any explicit supervision.

References

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